KNOWLEDGE / STUDY

NOV 9, 2026 · @ADI

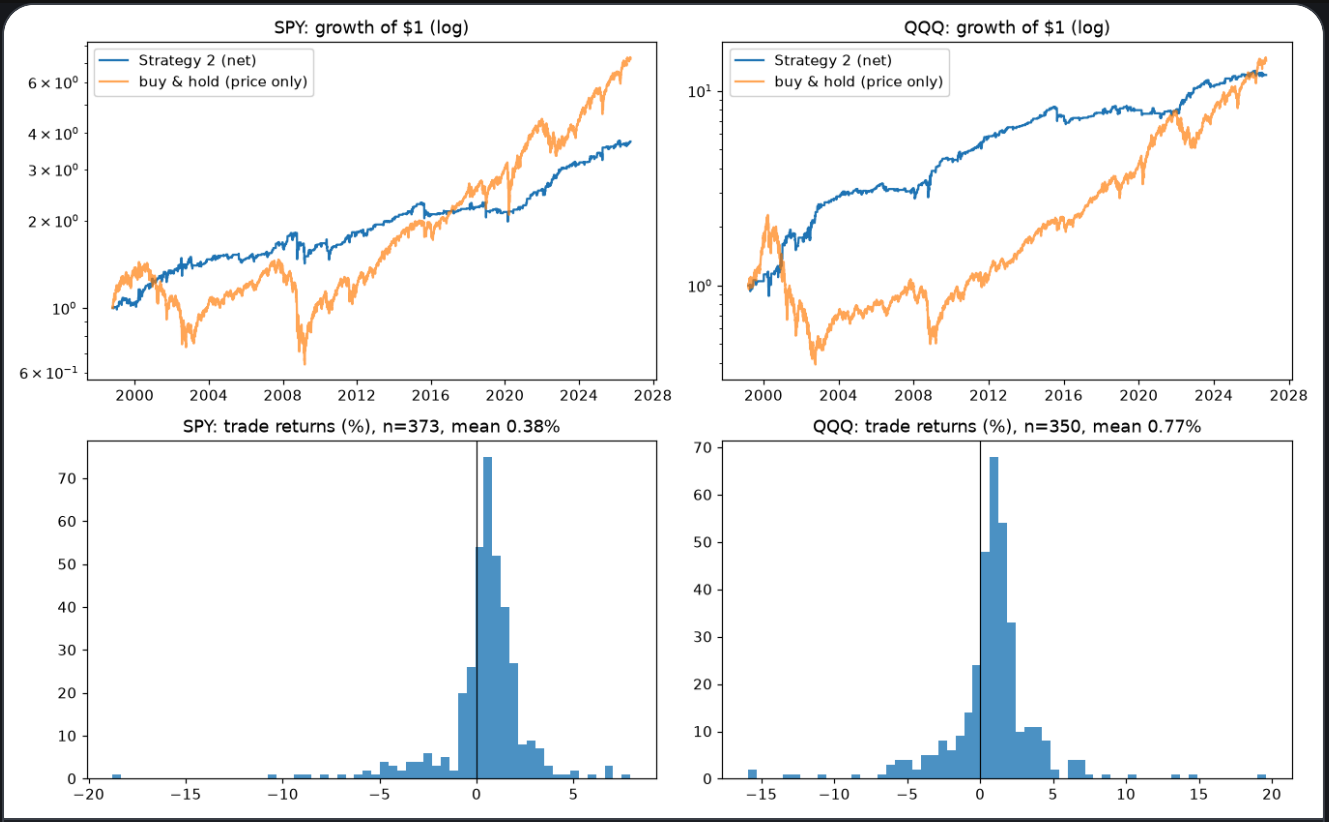
Backtest only. The figures below are a historical test of a rule. They are not a track record of Alphapick, not a forecast, and not investment advice.

At a glance

SPY and QQQ both won about three trades in four, and the strategy gave up far less in drawdowns than buy-and-hold, though it earned less in total.

MEASURE (OCT 1998 / MAR 1999 TO OCT 2026, NET OF COSTS)	SPY	QQQ
Trades (per year)	373 (13)	350 (13)
Win rate	73%	74%
Average trade	+0.38%	+0.77%
t-stat, and 90% range for the average	3.2, +0.18% to +0.57%	4.4, +0.48% to +1.06%
Average win / average loss	+1.26% / −2.03%	+2.00% / −2.78%
Profit factor	1.70	2.07
Average days held	3.2	3.2
Time in the market	17%	16%
Return per year: strategy / buy-and-hold	4.8% / 7.3%	9.4% / 10.1%
Worst drawdown: strategy / buy-and-hold	−21.6% / −56.5%	−25.1% / −83.0%
Same exit, entered on any day (control)	+0.04% per trade	+0.04% per trade

The strategy sits in cash 83% to 84% of the time, and these figures count that cash as earning nothing. Prices exclude dividends.



Top: growth of $1 on a log scale, net of costs, against buy-and-hold (price only). Bottom: how individual trade returns are spread, with most small gains and a thin tail of large losses.

The rule

Buy after three lower closes in a row, sell on the first close above the previous day's high, and cut the loss at three average daily ranges below the entry.

1. **Signal:** at the close, each of the last three closes is lower than the one before it.
2. **Entry:** buy at the next day's open.
3. **Exit:** sell at the close of the first day (the entry day counts) that closes above the previous day's high.
4. **Stop:** the entry open minus 3 times the 14-day average true range (ATR), measured on the signal day. If a day's low touches it, exit at the stop, or at the open if the market gaps below it. The average stop sits 4.5% below the entry on SPY and 7.1% on QQQ.

5 **Position:** long only, one position at a time. A new signal can be taken at the close of the exit day.

The entry matters, not just the exit. Buying on any day with the same exit and stop earns only +0.04% per trade, against +0.38% (SPY) and +0.77% (QQQ) after three lower closes. Per day held, the signal earns about 8 times as much on SPY (0.119% vs 0.014%) and 16 times as much on QQQ (0.241% vs 0.015%).

Stability over time

SPY's average trade has been steady for 28 years, while QQQ's was much stronger before 2013.

PERIOD	SPY TRADES	SPY AVERAGE	SPY T-STAT	QQQ TRADES	QQQ AVE
Through 2012	200	+0.37%	1.9	191	+1
2013 onward	173	+0.40%	3.1	159	+0
2024 onward	38	+0.48%	1.9	32	+0

SPY lost money in 5 of 29 calendar years (2003, 2005, 2015, 2018, 2019). QQQ lost money in 6 of 28 (2006, 2015, 2016, 2019, 2020, and the partial year 2026). QQQ's best years were the dot-com bust, 2000 to 2002, at about +1.9% to +2.2% per trade, which is why its long-run average is higher than its recent one.

Where it works best

The edge is largest in corrections: when price is below its 200-day average and fear is high, the bounce is bigger.

CONDITION AT THE SIGNAL	SPY: TRADES, AVERAGE	QQQ: TRADES, AVERAGE

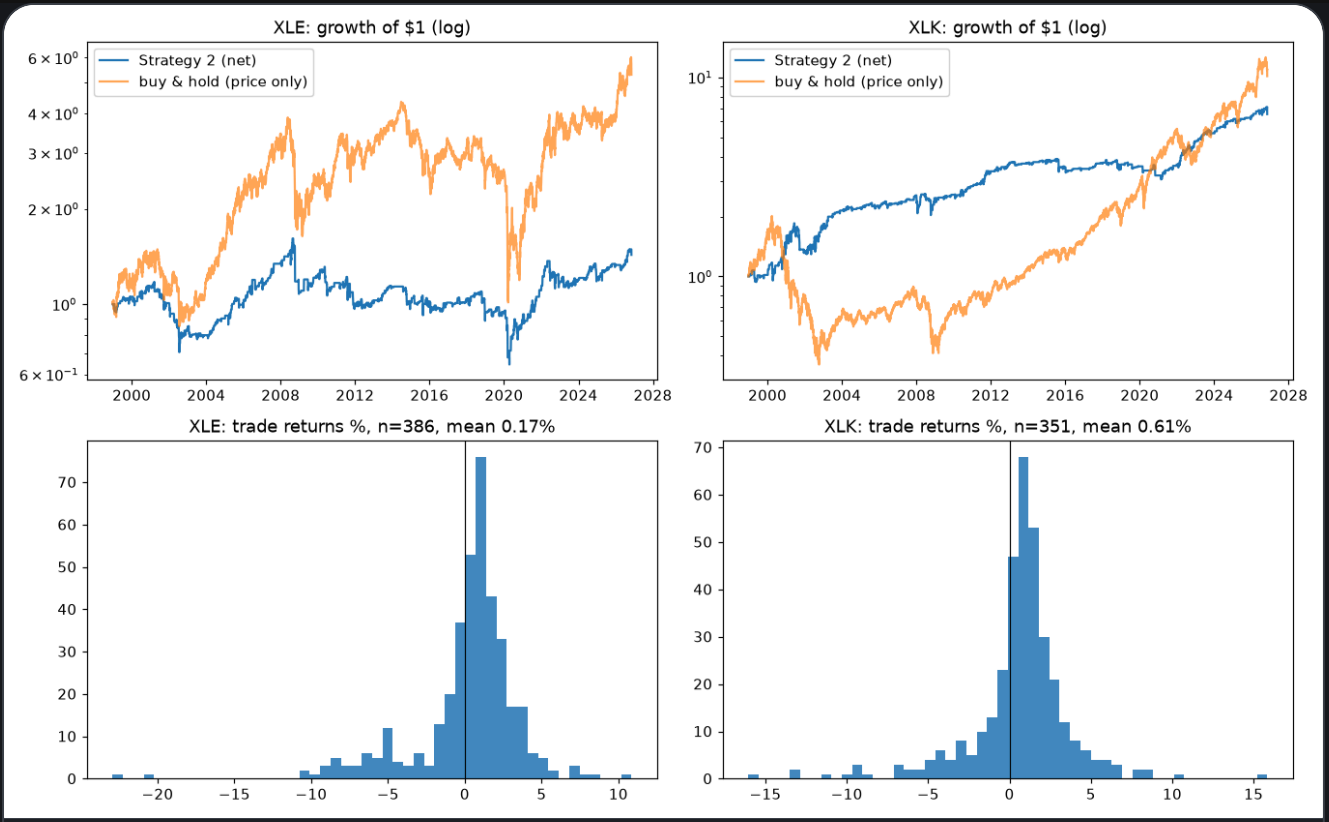
CONDITION AT THE SIGNAL	SPY: TRADES, AVERAGE	QQQ: TRADES, AVERAGE
Price below its 200-day average	134, +0.63%	127, +1.63% (t 4.4)
Price above its 200-day average	229, +0.23%	217, +0.26%
VIX above 30	34, +0.81%	35, +2.23% (t 3.2)
VIX 15 to 20	85, +0.22%	87, +0.22%
Three-day drop of 3% to 5%	85, +0.80% (t 3.2)	89, +0.84% (t 3.1)
Three-day drop of more than 5%	36, +0.38%	92, +1.55% (t 2.8)
Entry gaps down from the signal close	128, +0.60%	141, +1.23%
Entry gaps up	245, +0.27%	209, +0.46%

Some of this is a larger sample doing the work in one bucket, so read the pattern, not each number. Day-of-week differences (for example, QQQ trades entered on a Friday lost 0.05%) looked like noise given how many buckets were checked, and are left out of the table.

Other ETFs

Across 138 ETFs the rule had a positive average trade on 112 (81%) and was statistically significant (t-stat above 2) on 43, nearly all of them US stock funds. The full list, fund by fund, is in the next section; this one looks at the two sectors that differ the most.

XLK and XLE show how different two sectors can be. XLK averaged +0.61% per trade after costs (t-stat 3.6, profit factor 1.77) and was positive in 22 of 28 years. XLE averaged only +0.17% (t-stat 0.9, 90% range −0.12% to +0.45%), earned 1.3% a year against 6.2% for buy-and-hold, and its equity curve fell as much as 60% from a peak. XLE only worked when it was above its 200-day average (+0.51% per trade, against −0.26% below it).



The same two views for XLE (left) and XLK (right), through 5 Nov 2026.

ETF-by-ETF results

The dip-buy effect shows up across US stock funds and in long-term Treasuries, and not in commodities, currencies or most other bonds: 112 of 138 ETFs had a positive average trade and 43 had a t-stat above 2.

CATEGORY	ETFS	WITH A POSITIVE AVERAGE	T-STAT ABOVE 2	A
Broad US	14	14	9	
Style and factor	15	15	6	
US sectors	17	17	6	
Industries and themes	25	18	6	
International developed	18	17	1	

CATEGORY	ETFS	WITH A POSITIVE AVERAGE	T-STAT ABOVE 2	A
Emerging markets	8	8	2	
Bonds	15	8	4	
Commodities and currencies	14	3	0	
Leveraged (2x/3x)	12	12	9	
All	138	112	43	

How to read the tables below. Each table lists one category, best t-stat first. "Avg trade" is after costs, "Per year" compares the rule's return with buy-and-hold over the same years, and "fell X% at worst" is the rule's own deepest drawdown. The "Read" column grades the t-stat: Strong (3.5 or more), Good (2.5 to 3.5), Moderate (2 to 2.5), Weak (1 to 2), No edge (−1 to 1) and Negative (below −1).

Not 43 independent results. By chance alone, about three of 138 funds would reach a t-stat of 2, so 43 is far more than luck. But many of these funds hold the same stocks, so the number of truly separate confirmations is much smaller.

The effect is fading. Across the 127 ETFs with history on both sides, the average trade fell from +0.36% through 2012 to +0.20% from 2013 onward.

Costs end the story for quiet funds. Short-term Treasuries and municipal bonds (SHY, MUB) move so little each day that 2 basis points per side exceeds their normal range, so they lose.

Leveraged funds earn more per trade but fall further. Their average trades are +0.6% to +1.9%, with drawdowns of 36% to 85% and a worst single trade of −59% (SOXL). Inverse funds were not tested.

Different data from the sections above. These runs use Yahoo Finance prices adjusted for dividends, and 2 basis points per side, so figures for SPY (+0.41%), QQQ (+0.81%), XLK (+0.63%) and XLE (+0.21%) differ slightly from those earlier.

Broad US

All 14 broad US funds had a positive average trade (+0.34% on average) and nine were significant, with the Nasdaq-100 and S&P 500 funds strongest and small and mid caps weaker.

ETF	SINCE	TRADES	WIN	AVG TRADE	T	PE
QQQ · Nasdaq-100	1999	347	76%	+0.81%	4.8	
SPY · S&P 500 (SPDR)	1993	440	74%	+0.41%	3.8	
IVV · S&P 500 (iShares)	2000	343	75%	+0.39%	3.4	
VOO · S&P 500 (Vanguard)	2010	204	72%	+0.39%	3.4	
IWB · Russell 1000	2000	337	74%	+0.41%	3.4	
IWV · Russell 3000	2000	346	73%	+0.32%	2.6	
OEF · S&P 100	2000	316	71%	+0.30%	2.4	
IJR · S&P SmallCap 600	2000	351	75%	+0.35%	2.3	
VTI · Total US market	2001	316	70%	+0.27%	2.1	
IWM · Russell 2000 small caps	2000	349	71%	+0.27%	1.8	
MDY · S&P MidCap 400	1995	423	70%	+0.23%	1.7	

ETF	SINCE	TRADES	WIN	AVG TRADE	T	PER
DIA · Dow Jones 30	1998	370	68%	+0.18%	1.6	
RSP · S&P 500 equal weight	2003	289	70%	+0.24%	1.6	
IJH · S&P MidCap 400 (iShares)	2000	360	71%	+0.17%	1.1	

Style and factor

All 15 style funds were positive (+0.26% on average), with growth and momentum funds ahead of value and dividend funds, and three dividend funds (SDY, DVY, SCHD) losing money over the last three years.

ETF	SINCE	TRADES	WIN	AVG TRADE	T	PER YEA
VUG · Vanguard Growth	2004	275	72%	+0.39%	2.7	
IVW · S&P 500 Growth	2000	330	73%	+0.33%	2.5	
MTUM · Momentum factor	2013	169	75%	+0.41%	2.4	
USMV · Minimum volatility	2011	171	74%	+0.24%	2.3	
IWN · Russell 2000 Value	2000	346	71%	+0.34%	2.1	
QUAL · Quality factor	2013	168	74%	+0.31%	2.1	
VTV · Vanguard Value	2004	271	69%	+0.28%	2.0	

ETF	SINCE	TRADES	WIN	AVG TRADE	T	PER YEA
IWF · Russell 1000 Growth	2000	334	70%	+0.26%	1.9	
IWD · Russell 1000 Value	2000	327	67%	+0.23%	1.6	
SDY · Dividend aristocrats	2005	261	71%	+0.23%	1.6	
DVY · Dividend (select)	2003	282	74%	+0.24%	1.5	
IVE · S&P 500 Value	2000	319	65%	+0.20%	1.4	
IWO · Russell 2000 Growth	2000	354	72%	+0.22%	1.3	
VIG · Dividend growth	2006	245	76%	+0.17%	1.2	
SCHD · Dividend equity	2011	175	64%	+0.03%	0.2	

US sectors

All 17 sector funds were positive (+0.28% on average); technology and financials were the strongest, and utilities, materials, energy and communications the weakest.

ETF	SINCE	TRADES	WIN	AVG TRADE	T	PER
VGT · Vanguard technology	2004	290	75%	+0.67%	5.5	
XLK · Technology	1998	349	75%	+0.63%	3.7	
XLF · Financials	1998	364	72%	+0.51%	3.1	
XLP · Consumer staples	1998	325	74%	+0.28%	2.5	
XLV · Health care	1998	367	70%	+0.26%	2.3	
VNQ · Real estate (Vanguard)	2004	284	71%	+0.37%	2.1	
XLY · Consumer discretionary	1998	391	72%	+0.24%	1.8	
XLI · Industrials	1998	346	72%	+0.26%	1.8	
VFH · Vanguard financials	2004	293	71%	+0.31%	1.6	
IYR · Real estate (iShares)	2000	334	69%	+0.21%	1.3	

ETF	SINCE	TRADES	WIN	AVG TRADE	T	PER
XLE · Energy	1998	380	69%	+0.21%	1.2	
XLRE · Real estate	2015	141	65%	+0.16%	1.0	
VHT · Vanguard health care	2004	304	72%	+0.13%	1.0	
XLU · Utilities	1998	348	71%	+0.15%	1.0	
XLC · Communication services	2018	110	69%	+0.15%	0.7	
VDE · Vanguard energy	2004	310	69%	+0.12%	0.5	
XLB · Materials	1998	373	66%	+0.04%	0.3	

Industries and themes

18 of 25 industry and theme funds were positive (+0.16% on average); software, banks, semiconductors and retail worked, while energy, solar and uranium funds lost.

ETF	SINCE	TRADES	WIN	AVG TRADE	T
IGV · Software	2001	337	72%	+0.60%	4.0
KRE · Regional banks	2006	286	69%	+0.66%	3.4
SMH · Semiconductors	2000	374	68%	+0.59%	2.9
XRT · Retail	2006	281	74%	+0.49%	2.6
KBE · Banks	2005	293	69%	+0.53%	2.5
SOXX · Semiconductors (iShares)	2001	351	69%	+0.48%	2.5
XSD · Semiconductors (equal weight)	2006	277	70%	+0.31%	1.5
IYT · Transportation	2004	316	69%	+0.21%	1.4
GDX · Gold miners	2006	286	63%	+0.35%	1.2
ARKK · ARK Innovation	2014	165	69%	+0.37%	1.1

ETF	SINCE	TRADES	WIN	AVG TRADE	T
XHB · Homebuilders	2006	297	69%	+0.21%	0.9
ITB · Home construction	2006	273	69%	+0.22%	0.9
XBI · Biotech (equal weight)	2006	277	69%	+0.18%	0.8
JETS · Airlines	2015	167	69%	+0.21%	0.7
GDXJ · Junior gold miners	2009	244	62%	+0.26%	0.7
XME · Metals and mining	2006	284	64%	+0.12%	0.5
IBB · Biotech (Nasdaq)	2001	362	67%	+0.07%	0.4

ETF	SINCE	TRADES	WIN	AVG TRADE	T
ITA · Aerospace and defense	2006	275	69%	+0.06%	0.4
XOP · Oil and gas exploration	2006	293	66%	−0.10%	−0.3
LIT · Lithium and batteries	2010	244	66%	−0.09%	−0.5
ICLN · Clean energy	2008	258	66%	−0.14%	−0.6
OIH · Oil services	2001	369	62%	−0.22%	−0.9
PBW · Clean energy (Invesco)	2005	306	63%	−0.35%	−1.3
TAN · Solar	2008	282	61%	−0.45%	−1.5
URA · Uranium	2010	252	56%	−0.48%	−2.0

International developed

17 of 18 country and region funds were positive, but the average is small (+0.16%) and only Japan reached a t-stat above 2; the edge is weaker abroad than in

US stocks.

ETF	SINCE	TRADES	WIN	AVG TRADE	T
EWJ · Japan	1996	388	70%	+0.24%	2.2
EWS · Singapore	1996	364	67%	+0.30%	1.9
EWA · Australia	1996	367	72%	+0.25%	1.8
EWU · United Kingdom	1996	364	69%	+0.19%	1.4
EWP · Spain	1996	378	68%	+0.21%	1.4
EZU · Eurozone	2000	347	66%	+0.21%	1.4
VGK · Europe (Vanguard)	2005	279	68%	+0.21%	1.4
EWT · Taiwan	2000	363	67%	+0.16%	1.1
EWG · Germany	1996	394	66%	+0.15%	1.1
EWY · South Korea	2000	370	65%	+0.17%	1.0
EWI · Italy	1996	380	70%	+0.14%	1.0

ETF	SINCE	TRADES	WIN	AVG TRADE	T
EFA · EAFE developed markets	2001	319	66%	+0.11%	0.9
EWW · Mexico	1996	426	66%	+0.15%	0.8
EWQ · France	1996	400	65%	+0.10%	0.7
EWC · Canada	1996	361	69%	+0.09%	0.6
EWZ · Brazil	2000	356	67%	+0.12%	0.6
VEA · Developed markets (Vanguard)	2007	242	67%	+0.05%	0.3
EWH · Hong Kong	1996	402	61%	−0.02%	−0.1

Emerging markets

All 8 emerging-market funds were positive (+0.17% on average), with South Africa and China A-shares significant and the large broad funds (EEM, VWO) not.

ETF	SINCE	TRADES	WIN	AVG TRADE	T
EZA · South Africa	2003	324	70%	+0.48%	3.0
ASHR · China A-shares	2013	198	68%	+0.30%	2.3

ETF	SINCE	TRADES	WIN	AVG TRADE	T
EEM · Emerging markets (iShares)	2003	313	67%	+0.20%	1.6
INDA	2012	211	64%	+0.14%	1.2
FXI · China large caps	2004	322	66%	+0.18%	1.0
VWO · Emerging markets (Vanguard)	2005	279	66%	+0.06%	0.5
MCHI · China (MSCI)	2011	241	62%	+0.03%	0.2
THD · Thailand	2008	255	62%	+0.01%	0.1

Bonds

Only longer-term Treasury funds worked (TLT t-stat 3.8, TLH 2.9, IEF 2.2); corporate, high-yield and municipal funds did not, and short-term and municipal bonds lose to costs.

ETF	SINCE	TRADES	WIN	AVG TRADE	T
TLT · Treasuries 20+ years	2002	328	73%	+0.26%	3.8
TLH · Treasuries 10-20 years	2007	276	73%	+0.16%	2.9
IEF · Treasuries 7-10 years	2002	317	66%	+0.07%	2.2
BND · Total bond market	2007	245	67%	+0.07%	2.1

ETF	SINCE	TRADES	WIN	AVG TRADE	T
TIP · Inflation-protected Treasuries	2003	310	68%	+0.07%	1.8
AGG · US aggregate bonds	2003	290	66%	+0.04%	1.4
MBB · Mortgage-backed	2007	230	65%	+0.03%	1.0
IEI · Treasuries 3-7 years	2007	265	62%	+0.00%	0.1
BNDX · International bonds	2013	146	56%	−0.02%	−0.5
HYG · High-yield corporates	2007	275	63%	−0.06%	−0.5
LQD · Investment-grade corporates	2002	295	63%	−0.07%	−1.0
JNK · High-yield (SPDR)	2007	268	65%	−0.09%	−1.0
EMB · Emerging-market bonds	2007	277	58%	−0.09%	−1.2
SHY · Treasuries 1-3 years	2002	229	46%	−0.02%	−2.2
MUB · Municipal bonds	2007	235	46%	−0.23%	−4.7

Commodities and currencies

Only 3 of 14 were positive and none was significant: the rule did not work on gold, silver, oil, other commodities or currencies.

ETF	SINCE	TRADES	WIN	AVG TRADE	T	PI
UUP · US dollar index	2007	255	69%	+0.02%	0.4	
IAU · Gold (iShares)	2005	257	64%	+0.03%	0.2	
FXB · British pound	2006	327	63%	+0.00%	0.0	
GLD · Gold (SPDR)	2004	274	64%	−0.01%	−0.1	
CPER · Copper	2011	183	62%	−0.07%	−0.3	
FXE · Euro	2005	311	59%	−0.02%	−0.4	
FXA · Australian dollar	2006	280	66%	−0.03%	−0.4	
UNG · Natural gas	2007	311	64%	−0.12%	−0.4	
DBA · Agriculture	2007	283	65%	−0.04%	−0.4	
PDBC · Diversified commodities	2014	152	60%	−0.13%	−0.7	
FXY · Japanese yen	2007	309	63%	−0.05%	−1.0	

ETF	SINCE	TRADES	WIN	AVG TRADE	T	PI
DBC · Broad commodities	2006	287	65%	−0.15%	−1.1	
SLV · Silver	2006	245	58%	−0.36%	−1.5	
USO · Crude oil	2006	271	59%	−0.49%	−1.6	

Leveraged (2x and 3x)

All 12 leveraged funds were positive and 9 were significant, with average trades of +0.6% to +1.9%, but each carries large drawdowns, and TNA and SOXL fell 78% and 85% at worst.

ETF	SINCE	TRADES	WIN	AVG TRADE	T	PER Y
TECL · Technology 3x	2008	224	79%	+1.85%	4.3	
TQQQ · Nasdaq-100 3x	2010	212	76%	+1.71%	4.2	
QLD · Nasdaq-100 2x	2006	251	73%	+1.16%	3.7	
UPRO · S&P 500 3x	2009	220	73%	+1.30%	3.6	
FAS · Financials 3x	2008	219	74%	+1.49%	3.5	

ETF	SINCE	TRADES	WIN	AVG TRADE	T	PER Y
SPXL · S&P 500 3x (Direxion)	2008	232	72%	+1.24%	3.3	
TMF · Treasuries 20+ 3x	2009	239	71%	+0.70%	2.8	
SSO · S&P 500 2x	2006	254	73%	+0.78%	2.8	
TNA · Russell 2000 3x	2008	236	73%	+1.31%	2.6	
UWM · Russell 2000 2x	2007	260	72%	+0.71%	1.9	
UDOW · Dow 3x	2010	216	70%	+0.62%	1.8	
SOXL · Semiconductors 3x	2010	215	69%	+1.18%	1.7	

Risks

The rule wins small and loses large: every one of its worst trades was a stop-out, and they came in market crashes.

FUND	ENTRY DATE	LOSS	DAYS HELD	STOP DISTANCE
SPY	7 Oct 2008	−18.8%	4	16.3%
SPY	18 Feb 2009	−10.3%	9	10.3%
QQQ	31 Mar 2000	−15.9%	3	15.9%
QQQ	6 Oct 2008	−15.8%	5	15.7%
QQQ	31 Aug 2001	−13.3%	7	12.4%
XLK	6 Oct 2008	−16.0%	5	16.0%
XLE	6 Oct 2008	−22.9%	5	21.9%
XLE	10 Mar 2020	−20.9%	4	20.8%

The stop is three average daily ranges away, so it is wide exactly when volatility is high, and a gap can take the loss past it. Stop-outs are 7% to 11% of trades but average −5% to −7%, which cancels a large part of the gains from the winners.

Strategy drawdowns were shallower than buy-and-hold: −21.6% on SPY (Aug 2008 to Mar 2009) and −25.1% on QQQ (Mar to Apr 2000), against −56.5% and −83.0%.

A trade that is already losing rarely recovers. In the XLK and XLE tests, trades whose worst close was 4% or more below the entry finished positive only 8% (XLK, 38 trades) and 11% (XLE, 64 trades) of the time, and about half hit the stop. Only about one winning trade in ten was ever down 2% or more at a close, so the rule depends on the bounce coming almost at once. Because stops sit 5% to 9% away, size each trade by its stop distance, not by a fixed amount.

Robustness and costs

The result does not hinge on one setting: two to four lower closes and stops from 2 to 4 times ATR (or no stop) all stay positive.

VARIATION	SPY: AVERAGE (T-STAT)	QQQ: AVERAGE (T-STAT)
Base case: 3 lower closes, 3× ATR stop	+0.38% (3.2)	+0.77% (4.4)
2 lower closes	+0.28% (3.4)	+0.36% (3.2)
4 lower closes	+0.62% (3.2)	+0.83% (3.4)
Stop at 2× ATR	+0.36% (3.0)	+0.59% (3.1)
Stop at 4× ATR	+0.37% (2.8)	+0.82% (4.9)
No stop	+0.38% (3.7)	+0.90% (5.6)
Exit above the 5-day average instead	+0.29% (2.4)	+0.64% (3.6)
Exit on any up close instead	+0.19% (2.1)	+0.43% (3.3)
Exit after two days instead	+0.11% (1.2)	+0.33% (2.3)

The stop did not add return in this sample: with no stop the averages are the same or higher, and the worst drawdowns are about the same. It limits how far a single trade can fall, not how much the strategy makes.

Costs are not the problem. The average SPY trade falls from +0.41% with no costs to +0.38% at 1.5 basis points (0.015%) per side, +0.21% at 10 basis points and about zero at 20. QQQ goes from +0.80% to +0.77% at 1.5 basis points and +0.40% at 20. The rule also decides on the closing price, which cannot be traded exactly. Using the 15:55 price for both the signal and the exit changed the 2010 to 2026 average by only 0.03 points (SPY +0.44% instead of +0.47%, QQQ +0.49% instead of +0.52%).

Limits

A backtest cannot show that the edge will last, and several points here could make live results worse than the table.

1. **Few trades.** About 13 a year, so a bad stretch can last years before the statistics recover. QQQ's average since 2013 is +0.41%, and since 2024 its t-stat is only 1.3.
2. **A rising market.** ETFs climbed strongly over 1998 to 2026, and buying dips works best in rising markets. The data has no long, grinding decline.
3. **Chosen settings.** The 3× ATR stop and the exit rule were picked after comparing variants. The flat results across settings suggest a plateau, but live results will probably be worse than the best row.
4. **Many tests.** Dozens of variants and 138 ETFs were examined, and many of the ETFs move together, so some good-looking results are expected by chance.
5. **Tail losses.** The worst trades lost 16% to 23%, with stops up to 22% away in a crash. Gaps can go through a stop.
6. **Idle cash and dividends.** Cash earns nothing here, dividends and taxes are ignored, so the true return of a combined plan depends on what the idle money does.
7. **Fills.** The test assumes trades at the next open and at the close, and half a cent to a cent of spread. That is realistic for SPY and QQQ and is optimistic for smaller sector ETFs.

This is a historical test of a rule, not investment advice or a forecast.

Data and method

Everything here comes from daily and 1-minute price data and one script per test, with no outside forecasts.

1. **Daily prices:** Interactive Brokers historical data for SPY (from Oct 1998), QQQ (from Mar 1999) and 14 other ETFs, to 14 Oct 2026. Unadjusted for dividends. About 40 corrupt SPY bars from 2004 to 2008 and an unadjusted IWM split were repaired; the headline results moved by about 0.01 points.

2. **Recent weeks and the 15:55 check:** Databento 1-minute bars (EQUS.MINI) and Interactive Brokers 1-minute bars (2010 to 2026).

3. **Fear and trend context:** the VIX index (2005 onward, so earlier trades have no VIX reading) and each fund's 200-day average.

4. **Costs:** 1.5 basis points (0.015%) of price per side, about the cost of a half-cent spread on SPY or QQQ.

5. **Statistics:** the t-stat is the average trade divided by its standard error. Ranges around averages come from 5,000 random resamples of the trades.

6. **Assumptions:** buys at the next open, sells at the close, one position at a time, and if a stop and an exit could both happen on a day the stop is assumed to hit first.

7. **ETF universe (138 funds):** Yahoo Finance daily prices adjusted for splits and dividends (so bond and income funds are not distorted by monthly payouts), with 2 basis points per side. Inverse funds and funds with under about six years of history were left out.

← ALL KNOWLEDGE